Trading, Intuition, and Return

Certified Round-Trip Proof Translation for First-Order Theories and Automated Provers

ATLAS-audited mathematical revision v2.0

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Original: 26 October 2026; revised: 5 November 2026

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Abstract

A first-order Hilbert presentation with modus ponens (MP) and universal generalization admits a consequence-equivalent MP-only presentation when generalization is subject to the usual finite-support condition. We make the extra axiom schemas and the support-sensitive proof transformation explicit, construct reverse translations, and distinguish equality of contextual consequence from literal proof identity. We then connect an MP-only conditional proof graph to certified obligation discharge and original-rule decoding. A finite ready-frontier antichain captures redundant obligation sets, not operational history ancestry. The result is mathematical and conditional on exact axiom recognition and checking; it does not establish a universal elimination of rules, an ATP speedup, or independent proof-assistant certification.

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1 Scope and executive statement

Fix an ordinary classical first-order Hilbert calculus F with implication, standard quantifier axioms, MP, and a universal-generalization rule restricted by the undischarged support of its premise proof. Let $F^\star$ be the MP-only calculus defined in Section 3. The central result is

$$\Gamma \vdash_F T \quad \Longleftrightarrow \quad \Gamma \vdash_{F^\star} T \tag{1}$$

for every admitted hypothesis bank Γ and target formula T, with finite effective translations under effective-syntax assumptions. This is an exact statement about first-order *consequence*; it does not assert that an ATP discovers any proof faster or that decoding returns the identical proof syntax.

2 Presentations, proof occurrences, and the round trip

A presentation $P = (L, A, R)$ comprises a language L, axiom-instance family A, and inference rules R with all actual side conditions. $\mathrm{Cn}_P(\Gamma)$ consists of the formulas having finite derivations from Γ. Every finite proof uses a finite *support* $\Delta \subseteq \Gamma$ of undischarged assumptions. An *annotated proof occurrence* records its WFF, ordered premise occurrences, rule/axiom tag, finite support, and provenance of any imported certificate. Two proof occurrences may have the same WFF and different premises or supports.

A proof is Γ-*scope-closed* when every import is accompanied by a valid derivation from assumptions already authorized by Γ. Inferences must be checked, not inferred from a picture. A candidate dependency graph is not a proof by itself.

Theorem 2.1 (Abstract proof-translation criterion)**.** *If presentations P, Q share judgments and admitted banks and there are total functions E and D mapping each accepted finite P-proof of $\Gamma \vdash T$ to an accepted finite Q-proof of the same judgment and conversely, then $\mathrm{Cn}_P(\Gamma) = \mathrm{Cn}_Q(\Gamma)$ for all admitted Γ. Effective functions on encoded proofs yield effective witness translations.*

Proof. If $T \in \mathrm{Cn}_P(\Gamma)$, take a finite P-proof and translate with E to witness $T \in \mathrm{Cn}_Q(\Gamma)$. Use D for the reverse inclusion. □

This does not require $D(E(\pi)) = \pi$. Step count, sharing, derivation shape, and operational cost may change.

3 The classical first-order MP normalization

3.1 Precise input calculus

Take an ordinary capture-avoiding first-order language L. Let A be a standard classical Hilbert axiom-instance stock including a complete propositional implicational basis and standard quantifier axioms, notably the distribution theorem

$$\forall x(\alpha \to \beta) \to (\forall x\alpha \to \forall x\beta). \tag{2}$$

The standard substitution axioms include their usual free-for side conditions. Optional equality axioms and a specified additional nonlogical stock A_{extra} may be included. Every $a \in A$ is a zero-support axiom, and every such axiom occurrence can be generalized by the original rule under no open assumptions. For effective translation, axiom-instance membership and its intended expansion templates must be effectively recognizable.

The only proper rules of F are MP and

$$\frac{\varphi}{\forall x \varphi}\ \text{Gen} \qquad \text{only when } x \notin \mathrm{FV}(\Delta),$$

where Δ is the *actual undischarged support* of the derivation of φ at that occurrence. The formulation is intentionally support-sensitive: the finite proof of φ need not use every formula written in the ambient bank Γ.

For an ordered variable word $s = (x_1, \dots, x_n)$, including the empty word $()$, put

$$U_s\varphi = \forall x_1 \cdots \forall x_n \varphi, \qquad U_{()}\varphi = \varphi.$$

The order is syntactic, even when quantifier permutations happen to be provable. $\mathrm{Var}(s)$ denotes the set of variables appearing in s.

3.2 The MP-only axiom families

Define $F^\star$ with *MP as its sole proper inference rule* and the following exact stocks, for all indicated parameters:

$$(A) \quad A_s(a) := U_s a, \qquad a \in A; \tag{3}$$

$$(K) \quad K_s(\alpha, \beta) := U_s(\alpha \to \beta) \to \big(U_s\alpha \to U_s\beta\big); \tag{4}$$

$$(V) \quad V_s(\theta) := \theta \to U_s\theta, \qquad \mathrm{Var}(s) \cap \mathrm{FV}(\theta) = \varnothing. \tag{5}$$

The (V) freshness constraint is substantive. In general, $P(x) \to \forall x P(x)$ is not valid and is not an instance of (V).

Lemma 3.1 (The new axiom instances are original closed theorems)**.** *Every displayed (A), (K), or (V) instance is derivable in F from no open assumptions. Under the effective-presentation assumptions, a corresponding finite F-proof can be built algorithmically.*

Proof. For (A), begin with the axiom a, and generalize over $x_n, \dots, x_1$ in reverse order, producing $\forall x_1 \cdots \forall x_n a$. All steps are legal because support is empty.

For (V), begin a temporary derivation under the single assumption θ. The variables of s are all absent from $\mathrm{FV}(\theta)$, so successive universal generalizations in reverse order are permitted. The support-sensitive first-order deduction theorem then yields $\vdash_F \theta \to U_s\theta$. It is essential that the deduction theorem only discharges an assumption whose free variables are not generalized over in the relevant segment.

For (K), induct on the word length of s. The empty case is an implicational tautology. Let $s = (x, t)$ and suppose the result holds for t. Generalize that *closed theorem* over x:

$$\forall x\big(U_t(\alpha \to \beta) \to (U_t\alpha \to U_t\beta)\big).$$

Use Equation (2) first to pass from $\forall x U_t(\alpha \to \beta)$ to $\forall x(U_t\alpha \to U_t\beta)$, and then again to pass to $\forall x U_t\alpha \to \forall x U_t\beta$. Standard propositional composition yields the (K) instance for (x, t). This argument also works for repeated variables, because no fresh-variable substitution is invoked in the distribution instances. □

Theorem 3.2 (Context-preserving MP normalization)**.** *For every admitted* Γ *and formula* T*,*

$$\Gamma \vdash_F T \iff \Gamma \vdash_{F^\star} T.$$

Each inclusion has a finite effective proof-producing translation, conditional on the stated effective axiom presentation and proof syntax.

Proof. Encode F *into* $F^\star$*.* Fix a finite F-derivation with occurrences φ_i and finite support $\Delta_i \subseteq \Gamma$. By induction in topological order, establish the strengthened invariant

$$\mathrm{Var}(s) \cap \mathrm{FV}(\Delta_i) = \varnothing \quad \Longrightarrow \quad \Delta_i \vdash_{F^\star} U_s\varphi_i \qquad \text{for every finite word } s. \tag{6}$$

There are four cases.

1. An original axiom a has empty support, and $U_s a$ is an (A) axiom.
2. A hypothesis γ has support $\{\gamma\}$; use γ, the allowed instance $V_s(\gamma)$, and one MP.
3. If β follows by MP from α and $\alpha \to \beta$, let Δ_i contain the union of premise supports. Every s admissible for Δ_i is admissible for both premises. The induction hypotheses give $U_s\alpha$ and $U_s(\alpha \to \beta)$, and (K) plus two MP steps yields $U_s\beta$.
4. If $\forall x\psi$ follows from ψ by Gen, then $x \notin \mathrm{FV}(\Delta_i)$. For any s admissible at this occurrence, the longer word (s, x) also avoids $\mathrm{FV}(\Delta_i)$; applying the induction hypothesis to the premise produces $U_{(s,x)}\psi = U_s(\forall x\psi)$. No new generalization rule is introduced in $F^\star$.

Take $s = ()$ for the final occurrence. Every actual finite invocation of this structural induction terminates; in an implementation, `Build(i,s)` uses the four cases and finite original proof indices.

Decode $F^\star$ *into* F*.* Traverse an accepted $F^\star$-proof. Leave its Γ-hypothesis leaves and MP steps unchanged, and expand each (A), (K), or (V) leaf using Lemma 3.1. No new open assumptions are introduced. The final expanded proof derives T from Γ in F. Both procedures preserve the exact target and admitted hypothesis bank but need not preserve proof identity, size, or runtime. □

Corollary 3.3 (First-order theory axioms)**.** *Theorem 3.2 remains valid with any effectively recognized nonlogical first-order axioms* A_{extra} *entered as zero-support axioms. In particular, a standard Hilbert presentation of an ordinary classical first-order set theory has a consequence-equivalent MP-only* presentation*. No claim of decidable set-theoretic theoremhood follows.*

3.3 Implementable translation contract

A finite proof checker should validate: (i) exact formula parsing, including α-renaming conventions and free-for substitution; (ii) whether (A), (K), (V) parameters produce legitimate instances; (iii) proof-occurrence premise order and open-assumption support; (iv) proof-tree or proof-DAG sharing permissions; and (v) any imported lemma's receipt and scope. The encoder is schematic rather than a supplied executable checker. The decoder calls the original checker after expanding all axiom leaves.

4 Examples and the unrestricted-trading barrier

4.1 Legal and illegal generalization

With $\Gamma = \{P(x)\}$, the original rule permits $P(x) \vdash_F \forall y P(x)$ because y is absent from the open support. In $F^\star$ there is the exact proof

$$P(x), \quad P(x) \to \forall y P(x)\ (V), \quad \forall y P(x)\ (MP).$$

But $P(x) \vdash \forall x P(x)$ is invalid in general: take a structure where P holds of the assigned value of x but not all elements. The purported (V) axiom $P(x) \to \forall x P(x)$ is therefore forbidden.

4.2 A genuine obstruction to universal rule elimination

Counterexample 4.1 (Non-internalizable contextual effect)**.** *Start with a classical propositional Hilbert calculus and add the single unusual rule p/q for distinct atoms. Under $\{p\}$ the extended system proves q, but with empty assumptions it has only propositional tautologies as theorems: the p/q rule cannot be invoked for the first time in a closed proof because p is not among earlier tautological consequences. Hence $p \to q$ is not a closed theorem.*

Suppose an ordinary MP-only presentation Q retained the usual implicational axioms and had exactly the same consequence relation at every *bank. Then $\{p\} \vdash_Q q$ would, by the standard MP-only deduction theorem, imply $\vdash_Q p \to q$. Equivalence at the empty bank contradicts the previous paragraph. Thus MP cannot normalize* every *arbitrary extra inference rule while preserving all contexts by merely adding ordinary axioms.*

The obstruction does not apply to the specifically constrained first-order Gen operation proved in Theorem 3.2.

5 Theoremgraph intuition and certified return

The source theoremgraph program distinguishes *candidate*, *conditional*, and *realized* objects. A proposed reverse graph of inference dependencies is a guide for searching; it is not an inference licensed by reversing MP. For MP,

$$\frac{P \qquad P \to T}{T} \qquad \rightsquigarrow \qquad \text{to obtain } T\text{, search for } \textit{both } P \text{ and } P \to T.$$

The arrow $\rightsquigarrow$ denotes a *planning instruction*, not a formal proof rule.

Definition 5.1 (Three statuses)**. Candidate** R_{cand}: an unverified reverse AND/OR suggestion object stored only in CAND. **Conditional** R_{cond}: a selected finite accepted $F^\star$ proof DAG $\pi^\star : \Gamma \cup O \vdash_{F^\star} T$, with a finite set O of explicitly open formula obligations, valid ordered MP ports, authorized imports, and scope annotations. **Realized** R_{real}: each obligation has a compatible Γ-proof; after replacing every open leaf and checking the assembled proof, no non-Γ assumption remains.

Theorem 5.2 (Trading–theoremgraph–return composition)**.** *Let $F, F^\star, \Gamma, T$ be as in Theorem 3.2. Suppose a finite conditional derivation $\pi^\star : \Gamma \cup O \vdash_{F^\star} T$ is accepted by an exact $F^\star$ verifier, every imported lemma is scope-closed over Γ, each $o \in O$ has an independently accepted $\Gamma \vdash_{F^\star} o$ proof, and the expanded decoder's output is accepted by the final original checker. Then $\Gamma \vdash_F T$. Conversely, any accepted original proof can be encoded as a realized graph with $O = \varnothing$.*

Proof. Replace each open obligation leaf by its accepted Γ-proof, retaining ordered MP premises, and expand all authorized imported proof certificates. Because the rule stock of $F^\star$ is MP-only and its new axiom leaves are closed theorems of F, this substitution creates no hidden generalization side-condition. The resulting finite derivation is checkable under precisely Γ. Apply the decoder from Theorem 3.2 and run the final original checker. For the converse, encode a fully closed F-proof; its reversed dependency DAG has no open obligations. □

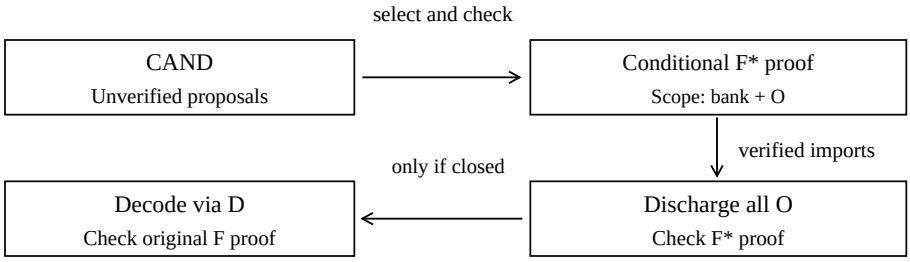


Figure 1: The certification gate. Reverse inference is a search proposal, never itself a proof shortcut.

If an OR node offers alternative derivations, checking one selected branch validates only that branch. Proofs carrying undischarged assumptions outside Γ must not be deposited into BANK.

6 Finite minimal obligation antichains and the SIC interface

Let $D_1, \ldots, D_m$ be finitely many independently checked conditional skeletons for the same target, with finite outstanding sets $O_1, \ldots, O_m$. For a certificate-backed admissible Γ-ledger B, define the readiness predicate

$$\Phi_T(B) := \bigvee_{i=1}^{m} [O_i \subseteq B].$$

Proposition 6.1 (Ready-frontier antichain)**.** *Deleting any listed O_j strictly containing a different listed O_i leaves Φ_T unchanged. The distinct inclusion-minimal sets form an inclusion antichain representing the same readiness region for this finite list. If $B \subseteq B'$ are compatible ledgers that preserve receipt validity, $\Phi_T(B) \leq \Phi_T(B')$.*

Proof. If $O_i \subsetneq O_j$, any ledger containing O_j also contains O_i, so the jth disjunct is redundant. Delete such sets until none remains. Readiness is monotone under a receipt-preserving extension because every previously satisfied subset test remains satisfied. □

For example, $\{L_1\}$, $\{L_2, L_3\}$, $\{L_1, L_4\}$ reduce to the first two minimal sets. This is *not* the same object as an antichain of incomparable operational SIC histories. A scheduler in an ATP or an AMLD-type settler may dispatch open obligations, but this formal proposition neither demonstrates a multi-agent implementation nor any parallel speedup. In particular B denotes a *compatible ledger*, not a silently mutable frozen input BANK.

7 Resource costs, experiments, and audit obligations

Consequence equivalence alone supplies no usable resource bound. A realistic charged cost decomposition is

$$W = W_{\text{encode}} + W_{\text{propose}} + W_{\text{check}} + W_{\text{search}} + W_{\text{import}} + W_{\text{compose}} + W_{\text{decode}} + W_{\text{final}}.$$

A count of MP in a composed DAG neglects discovery, lookup, proof-tree serialization, source/certificate custody, and final independent checking. An honest comparative experiment must freeze the target family, input Γ, verifier, theory, budget, and machine; compare against original-rule provers already allowed normal goal-directed search; and report UNKNOWN for exhausted budgets.

ID	Audit item	Revision status
A01	Specify first-order axioms, substitution, equality, and support-sensitive Gen.	Explicit assumptions
A02	Prove (K) for arbitrary ordered prefixes, including repetitions.	Expanded induction
A03	State the deduction-theorem hypothesis for (V).	Corrected
A04	Explain the all-prefix induction and Gen case.	Full proof
A05	Make decoding conditional on effective axiom recognition and verification.	Corrected
A06	Support unrestricted-trading counterexample.	Proof included
A07	Ensure proof DAG obligations/imports are Γ-closed before BANK admission.	Certification gate
A08	Separate global compilation from one-line local resource internalization.	Clarified
A09	Antichain represents only the finite listed skeletons.	Scoped
A10	Repair malformed figures and remove implication of independent certification.	Redrawn/qualified

The earlier text referred to "Theorem 3.1" where the statement was actually *Lemma 3.1*; the repaired document uses Lemma 3.1 consistently. No independent proof assistant was invoked for this revision.

8 Open research conjecture and release status

It remains open whether there is a natural infinite family of first-order proof tasks for which theoremgraph-guided MP-only search, *including* encoding, obligation search, decoding, and final verification, has provably lower fully charged settlement cost than a genuinely matched original-rule method. Also open is the stronger operational question of when translations preserve selected search states, budgets, and progress. Proof-theoretic equivalence does not by itself imply bisimulation or any stage-by-stage runtime guarantee.

References

1. B. Tenneson, *Trading Theorem 003: Mutual Reproducibility, Consequence-Preserving Trades, Quotient Hunting, and Verified Settlement Search*, audited edition, September 2026, `https://btenneson.github.io/pub/cs.LO_Logic_in_Computer_Science/trading_theorem_003/`.

2. B. Tenneson, *OPEN. AKM: Theorgraph Reversal and the Conversion of Intuition into Certified Proof*, revised working manuscript, October 2026. The current article uses the corrected term *theoremgraph*.

3. B. Tenneson, *CAMI's Theorems and Theories, Volume I: The Antichain Frontier*, author library, `https://btenneson.github.io/pub/cs.LO_Logic_in_Computer_Science/camis_theorems_and_theories_volume_i/`.

4. B. Tenneson, *Exact Addressing*, working revision 4.14, on annotated proof dependency graphs.

5. Standard first-order Hilbert logic: implication distribution, contextual generalization, and the support-sensitive deduction theorem are established background facts rather than claimed novel results here.

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